

Superstore Sales Analytics & Operational Forecasting

Executive Summary

This report summarizes an analysis of four years of Superstore sales data (2015-2018) to help guide inventory and revenue decisions. We built a forecasting model that predicts sales three months ahead with a good degree of accuracy, identified three unusual sales events worth investigating, and grouped our products into three demand categories, each requiring a different stocking approach. Overall, the business shows steady year-over-year growth driven by strong seasonal peaks around September through December, and the analysis points to three specific actions that can reduce stockouts, lower excess inventory, and protect revenue in the coming quarter.

Key Findings from Data Analysis & Forecasting

- Technology is the top revenue-generating category, with total sales of \$827,456 over the analysis period, ahead of Furniture and Office Supplies.
- The East region shows the most consistent year-over-year growth of any region; average delivery time across all regions is about 4 days.
- Sales are clearly seasonal, with September, November, and December consistently the strongest months, on top of a steady long-term upward trend.
- We tested three forecasting approaches (SARIMA, Prophet, and XGBoost). The tuned XGBoost model was the most accurate, with an average prediction error of about **\$13,900 per month (MAPE $\approx$ 14%)**, making it the best model for short-term sales forecasting.

• 3-Month Sales Forecast: January - March 2019

The table below shows projected sales by category and for the East region. Based on the model's historical performance (**MAPE $\approx$ 14%**), actual results are expected to fall within approximately $\pm 14\%$ of these forecasts under normal business conditions.

Month	Furniture	Technology	Office Supplies	East Region
Jan 2019	\$10,700	\$26,100	\$21,400	\$18,300
Feb 2019	\$6,050	\$24,000	\$28,200	\$19,900
Mar 2019	\$16,900	\$28,000	\$26,600	\$12,500

In plain terms: Furniture and Technology sales are both expected to grow through the quarter. The East region, by contrast, is projected to decline by roughly 32% from January to March, which is the single largest area of concern in this forecast.

Top 3 Sales Anomalies and Likely Causes

- Week of March 22, 2015 - An unusually large spike in sales, flagged by two independent detection methods. Likely cause: a major promotion, an early Easter holiday, or a large one-off corporate order.

- Week of January 4, 2015 - An unusually low sales dip. Likely cause: the typical post-holiday slowdown following the Christmas and New Year period.
- Week of December 17, 2017 - An unusually high sales spike. Likely cause: last-minute pre-Christmas shopping surge.

Product Demand Segmentation & Stocking Strategy

- Growing High-Value Products (Copiers): High order value, fastest growth, but volatile demand. Strategy: keep lean on-hand stock, monitor demand in real time, and use flexible supplier agreements rather than large standing inventory.
- High-Volume, Mature Products (6 sub-categories, incl. Chairs, Phones, Storage): Core, dependable revenue drivers with steady demand. Strategy: maintain consistently high stock levels with regular, predictable reordering to avoid stockouts.
- Low-Volume, Stable Products (10 sub-categories, incl. Art, Labels, Envelopes): Lower sales volume but very predictable. Strategy: lean, demand-based replenishment using standard reorder points to minimize holding costs.

Business Recommendations

- Launch a targeted East region campaign for Q1 2019. Data backing: the East region is forecast to decline about 32% (from roughly \$18,300 to \$12,500) between January and March 2019, the clearest downside risk in the forecast.
- Move to agile, vendor-managed inventory for Copiers. Data backing: Copiers combine the highest growth rate with the highest sales volatility of any product group, so both stockouts and overstock are costly under a standard fixed-order approach.
- Pre-build stock ahead of the Q4 holiday peak for high-volume, mature products. Data backing: anomaly detection confirmed sharp, repeatable demand spikes in mid-to-late December (e.g., the week of Dec 17, 2017), consistent with the strong seasonal pattern found in the broader trend analysis.

Risk & Limitation

This forecasting system is built entirely on historical sales patterns. It has no way to anticipate events with no precedent in the training data - such as a new competitor entering a region, a supply chain disruption, or a broader economic shift. If such an event occurs, forecast accuracy will drop until the model is retrained on new data. We recommend revisiting and recalibrating the model quarterly, or immediately after any major, unexpected market event.